

A REFLECTION ON THE ELECTRON

The Standing Wave Mechanisms and Spin Topology

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Abstract

In the previous article, “A Reflection on the Magnetic Field,” we elucidated the physical ontology of electromagnetic fields and waves as microscopic “Moiré patterns” on the computational grid. This paper extends this logical chain further into the microscopic realm, striking directly at the most fundamental unit of matter—the electron. Starting from the underlying propagation conditions of electromagnetic waves, we re-examine the true physical entity morphology of the electron within the atom, attempting to reveal the profound grid dynamical mechanisms behind its spin and magnetic moment. We propose that the electron is not a “point particle” in the traditional sense, but rather a macroscopic electromagnetic standing wave formed by “entropy momentum density fluctuations” (photons) on the underlying computational grid, which undergo path bending and self-trapping under their self-generated “momentum pressure.” By introducing the “beat delay” mechanism of cross-layer information and the environmental dependence of the effective speed of light, this model provides a purely mechanistic and grid-dynamical explanation for the ontology of charge, the equivalence of mass and charge, the radiation mechanism of quantum transitions, and the abstract 4π topological symmetry of spin.

1 Theoretical Reconstruction of Electromagnetic Wave Propagation Conditions

Electromagnetic waves are essentially the propagation of accelerated changes in entropy momentum density (corresponding to the classical magnetic vector potential) through space. As a photon propagates, it constantly generates electric field momentum in the direction perpendicular to its propagation. To accurately describe the interactions between different charges, we must stratify and refine the electric field effects at the $b_0 - b_1$ layer, and deepen and correct the initial assumptions regarding “entropy momentum” in the theory.

1.1 The Microscopic Essence of Entropy Momentum: “Beat Delay” and the Ontology of Charge

- **Cross-Layer Beat Delay:** Entropy momentum itself is a microscopic bit-timing pattern. We consider it to be a cross-computational-layer **beat delay mechanism**, manifested as a deterministic timing relationship between the b_0 and b_1 layers. That is, a pattern first appears on b_0 and is then copied to b_1 , or it first appears on b_1 and is then copied to b_0 .
- **Generation of Electric Field Momentum:** This intrinsic timing asymmetry cannot spontaneously reach equilibrium. Its propagation in different directions through space constitutes the **physical ontology of electric field momentum**.
- **Definition of Positive/Negative Charge:** The two opposite beat timings described above correspond respectively to the fields generated by two different charges. We define these as **positive charge momentum** and **negative charge momentum** (abbreviated as positive momentum and negative momentum). A charge is essentially a generator that continuously creates this “beat difference” between these layers.

1.2 The Differentiation of Photons and the Emergence of Absolute Light Speed

- **Positive Photons and Negative Photons:** Due to the different momentum generation mechanisms, the microscopic morphologies of the entropy momentum density fluctuations generated by positive and negative charges are vastly different. We distinguish them as **positive photons** and **negative photons**. Because of the spatiotemporal asymmetry of their massive microscopic information structures, two beams of opposite-type photons will not cancel each other out when they meet.
- **Macroscopic Equivalence:** Although their microscopic structures differ, if we only consider spatial relativistic effects, the essence of a photon is a wave; fluctuations of thrust and pull are equivalent in their effects, and the two can transform into each other during interactions with atomic nuclei. Meanwhile, the propagation and interaction of entropy momentum within the computational layer strictly follow the **principle of linear superposition**.
- **Environmental Dependence of Light Speed (Momentum Pressure Effect):** The speed of light is an emergent phenomenon of the underlying grid.

The accelerated variation of electric field momentum density propagates in the perpendicular direction. **Its effective propagation speed is constrained by the “momentum pressure” of the space it occupies.**

- **Homogeneous Deceleration:** Pre-existing negative momentum in space exerts a “pressure” on a negative photon propagating perpendicular to the momentum direction. This rigid pressure constitutes a propagation resistance, thereby exerting a **decelerating effect** on the negative photon. Furthermore, the effects of the same type of momentum moving toward each other will not cancel out, but will instead be strengthened.
- **Heterogeneous Acceleration:** Conversely, positive momentum in the environment reduces the propagation resistance for negative photons, **accelerating** their motion, though its upper limit remains constrained by the speed of light in a vacuum.

2 The Ontology of the Electron: A Self-Trapped Photon Standing Wave

Within the framework of “*Computational Realism*,” the physical entity of the electron cloud (probability cloud) itself is the entirety of the electron.

- **The Electron Entity as an Electromagnetic Standing Wave:** The electron is a macroscopic standing wave system composed of electromagnetic waves (photon groups) moving in circular motion (negative charges are composed of negative photons, positive charges of positive photons). The continuous existence of photons creates electric field entropy momentum, and the electric field entropy momentum in turn distorts the propagation paths of the photons. The two act as cause and effect for each other, maintaining an eternal cycle of photon entropy momentum.
- **Self-Trapping Mechanism Induced by “Refractive Index”:** Within the spatial region of the electron cloud, the circulating photons continuously “pump out” a fixed entropy momentum. Although the electric field vector intensity at the geometric center of the electron cloud is zero (consistent with QED models), its **electric field pressure density (a scalar) is not zero at the center; rather, it rises sharply the closer it gets to the center.**
- **Dynamics of Path Bending:** According to the aforementioned “momentum pressure” deceleration effect, the closer to the center, the lower the effective

propagation speed of the photon. This extreme spatial gradient of the speed of light forms a powerful “**refractive index**” on the grid, forcibly twisting the linear motion paths of the photons and compelling them into circular motion. However, the photon density does not increase infinitely; as its density increases, photons will gradually deflect from tangential motion toward the center. Due to geometric relationships, the central momentum pressure is always greater than that on the outside. Eventually, photons will penetrate the center and propagate outward again; this process limits the photon density.

3 The Scale Morphology and Dynamical Transitions of the Electron

The electron is a flexible dynamical structure deeply coupled with its environment. Its volume and morphology depend on the standing wave conditions achieved by the interaction between photons and the electric field.

- **The Boundary Line of Volume (Speed of Light Limit Constraint):** On the outer edge of the electron cloud, as the diameter increases, the electric field pressure density drops rapidly, and the speed of the outer circulating photons increases accordingly. When the peripheral speed of light approaches the absolute vacuum speed of light c , the speed encounters the physical bottleneck of the grid and cannot increase further. This physical limit destroys the proportional requirements for maintaining the standing wave, making it impossible for the volume of the electron entity to expand infinitely. **This limit boundary macroscopically determines the exact magnitude of the elementary charge e .**
- **Equivalence of Electron Mass and Charge:** The mass of an electron is always perfectly equivalent to its charge. Ontologically, both represent the same fact: exactly how many circulating photons are “bound” within this standing wave structure (these photons pump out momentum at a constant rate).
- **Environmental Dependence and Photon Emission (The Essence of Quantum Transitions):** The morphology of the electron cloud always represents the effective standing wave form that can be maintained under the current environment.
 - **Mechanism of Energy Level Transitions:** When a free electron falls into the potential well of an atomic nucleus, due to the accelerating ef-

fect of the nucleus's positive momentum environment on the electron's negative photon flow, the **effective speed of light** of the photons inside the electron **increases**. This directly leads to the collapse of the original standing wave proportional conditions; the outer structures of the electron cloud are stripped away, and the excess energy is radiated outward in the form of **free photons**.

- Correspondingly, in the electron orbitals outside the nucleus, due to geometric relationships, the closer a region is to the nucleus, the greater the spatial rate of change of the nucleus's positive electric field momentum. This causes the curvature of the electron cloud to become greater, thereby shaping the morphologies of electron clouds at different energy levels.

4 The Topological Code of Angular Momentum and Magnetic Moment

The abstract properties of spin and magnetic moment in the Standard Model possess intuitive explanations rooted in pure geometry and fluid topology under this dynamical framework.

- **The Dynamic Origin of Magnetic Moment:** Because the photons inside the electron cloud undergo high-speed periodic motion, the entropy momentum density they pump outward is not absolutely constant in its spatiotemporal distribution. This high-frequency periodic fluctuation in density directly constitutes the source of the macroscopic magnetic moment.
- **Spin Angular Momentum and Orbital Angular Momentum:** Essentially, these are an effective physical decomposition of the same macroscopic entity. The reality is that the topological phase cycles of the photons are acted upon by external electric field entropy momentum, causing a reorganization of their shape and scale.
- **The Topological Essence of Spin-1/2 (Dual Orthogonal Precession):** Within the entropy momentum phase of the electron standing wave, there actually exist **two mutually orthogonal, independent “precessional” cycles** (geometrically analogous to satellite orbits with a precession axis). To maintain the stability of the standing wave structure, the cycle periods of these two dimensions must remain absolutely consistent.

- **Constraints of the Dual Cycle and Lorentz Effects:** When an electron is acted upon by external electric field entropy momentum, differences in effective propagation speeds arise across different regions. To strictly satisfy the consistency requirement for the circulation speeds in these two orthogonal directions, the overall topological morphology of the electron must reorganize. This requirement for internal periodic synchronization is the root cause of the electron's length contraction effect. Furthermore, when the electron moves, the actual grid motion path of the photons becomes larger, allowing the standing wave mode to accommodate more mass.
- **Geometric Decoding of 4π Symmetry:** The fundamental reason why a spin-1/2 angular momentum must rotate 720° (4π) in space to return to its original state is that the phase standing wave morphology of its entropy momentum density is based on a dual orthogonal cycle on a spherical surface. A rotation of 360° (2π) in only a single plane cannot simultaneously reset the cycle of the other orthogonal phase.

- **The Dynamical Essence of the Pauli Exclusion Principle: Topological Meshing of Complementary Phases**

That an electron is a standing wave of photons means it must preserve the fluctuating state of momentum density. Two electrons with identical spins (co-directional 4π chiral precession) cannot coexist by meshing their momentum densities. However, when two electrons possess “opposite spins,” the two standing wave systems serve as **perfect mirrors** of each other across their dual orthogonal topologies. When they occupy the same orbital space, the dense and sparse regions of their photon momentum flows complement each other, and their directions of motion are opposite. This “perfect phase complementarity” causes the two high-frequency photon streams to act like perfectly engaged four-dimensional mechanical gears, interpenetrating in three-dimensional space without mutual interference. They share the same macroscopic volume, but their microscopic photon momentum flows are perfectly staggered in four-dimensional spatiotemporal coordinates, thereby avoiding the superimposition of “momentum pressure.”

This topological meshing directly triggers two decisive macroscopic physical consequences:

1. **Magnetic Moment Returns to Zero (Texture Dissolution):** Because the moving dense and sparse regions are perfectly leveled out by mirroring, the “microscopic Moiré patterns” bearing periodic fluctuations, which originally radiated outward individually, are internally can-

celed out at extremely close range. The system no longer leaks dynamic momentum texture residues to the outside, which means **the macroscopic magnetic moment of the composite system is flattened to 0.**

2. **Doubling of the Electrostatic Field:** Although the dynamic textures are canceled out, this does not erase the foundational computational fact that the two electrons serve as underlying “beat generators.” They continue to independently pump out negative momentum flows, thereby causing the outwardly apparent electrostatic pressure to exactly double ($-2e$).